SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I M. Arshini 192424932L *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Arshini

25 + 25 + 25 + 12 + 8

95

2

Annexure A

Constraint-Based Problem Solving

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1. A hospital needs to assign 3 nurses (N1, N2, N3) to 3 wards (ICU, General, Emergency) such that:

   Constraints:

   i. N1 cannot be assigned to Emergency ward

   ii. N2 must be assigned before N3 (ICU < General < Emergency)

   iii. Only one nurse per ward

   iv. N3 cannot be assigned to ICU

   v. All nurses must be assigned exactly one ward

   Apply Backtracking Search to find a valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

2. A cleaning robot operates in a 5x5 room-grid from Start (S) to a docking Goal (G), with a limited battery allowing at most 8 moves.

Constraints:

i. Movement allowed: Up, Down, Left, Right

ii. Cost of each move = 1 (battery unit)

iii. Cannot pass through furniture (obstacle) cells

iv. The robot must reach G within the battery limit

Using the above constraints and Depth-Limited Search (with the battery limit as depth bound), show step-by-step node expansion and determine whether a valid path exists within the limit, along with its cost.

Regno: - 1[illegible]

Hospital Nurse Assignment using Backtracking search

Variables

* Nurses : N1, N2, N3

Wards

* ICU
* General
* Emergency

constraints :-

N1 ≠ Emergency

N2 must be assigned before N3

Order : ICU < General < Emergency

one nurse per ward

Every nurse gets exactly one ward.

step 1 :- Assign N1

Try ICU

N1 → ICU

constraint check

N1 Not emergency

ward unique

current Assignment	ward
N1	ICU
N2	—
N3	—

Step 2:- Assign N2

Available wards:

General

Emergency

Try General

N2 → General

Constraint check

ward unique

N2 before N3

Current Assignment

Nurse	Ward
N1	ICU
N2	General
N3	

Step 3:- Assign N3 :-

Remaining ward

→ Emergency

Check constraints

N3 not ICU

ICU < General < Emergency

N2 = General

N3 = Emergency

General becomes before Emergency.

Assignment Accepted.

Search tree

start

- N1 → ICU
  - N2 → General
    - N3 → Emergency ✓
  - N2 → Emergency ✗

    N3 would have to take General, violating N2 before N3
- N1 → General
  - N2 ← ICU
    - N3 → Emergency ✓
  - N2 → Emergency? ✗
- N1 → Emergency ✗

  (constraint violated)

Backtracking table

step	Assign	constraint check	result
1	N 1 → ICU	N1 not emergency	✓
2	N 2 → General	unique ward	✓
3	N 3 → Emergency	N3 not ICU, N 2 before N3	✓

final valid schedule

Nurse	ward
N1	ICU
N2	General
N3	emergency

2) cleaning Robot using Depth limited search (Depth limit = 8)

Assume the following 5x5 grid.

S = start
G = Goal
X = obstacle
. = free call

	C1	C2	C3	C4	C5
R1	S	-	.	X	-
R2	X	X	.	X	.
R3	.	.	-	-	-
R4	-	X	X	X	.
R5	^	.	.	.	G

start = (1, 1)
Goal = (5, 5)
Depth limit = 8
move cost = 1

step 1 :-

current node :-

(1,1)

Depth = 0

Expand

possible moves

Right → (1,2) ✓

Down → obstacle ✗

visited

(1,1)

step 2 :-

current Node :-

(1,2)

Depth = 1

Expand

Right → (1,3) ✓

Left → Already visited

Down → obstacle ✗

visited

(1,1)

(1,2)

step 3 :- current node :-

(1,3)

Depth = 2

Expand

Down → (2,3)

Right → obstacle

left → visited

visited

(1,1)

(1,2)

(1,3)

<u>step 4</u>

current node:-

(2,3)

Depth = 3

Expand

~~Down~~ Right → (3,3) ✓

~~Right~~ Down → obstacle

visited

(3,4)

left → obstacle

visited

(1,1)

(1,2)

(1,3)

(2,3)

Step 5 :-

current node:-

(3,3)

Depth = 4

Expand

Right → (3,4)

left → (3,2)

choose right

visited

(3,3)

<u>step 6</u> :-

current node :-

(3,4)

Depth = 5

Expand

Right → (3,5)

Down → obstacle

(3,4)

Step 7 :-

current node

(3,5)

Depth = 6

Expand Down → (4,5)

visited (3,5)

Steps:-

[illegible]

([illegible] 5)

Depth = [illegible]

[illegible] d

[illegible] → (5,5) Goal

depth = 8

Goal found

Depth limited search tree

(1,1)
└ (1,2)
  └ (1,3)
    └ (2,3)
      └ (3,3)
        └ (3,4)
          └ (3,5)
            └ (4,5)
              └ (5,5) Goal

path found :- (1,1)
→ (1,2)
→ (1,3)
→ (2,3)
→ (3,3)
→ (3,4)
→ (3,5)
→ (4,5)
→ (5,5)

cost calculation

move	cost
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1

Total cost = 8 battery unit

RUBRICS FOR CALCULATION:

[illegible]	Weightage	[illegible]	[illegible]	[illegible]	[illegible]
Problem Understandin g	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

25+25+25+12+8

= 95

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